Cloud Volumes ONTAP

Hybrid Cloud Solutions

NetApp
December 11, 2020

This PDF was generated from https://docs.netapp.com/us-en/hybridcloudsolutions/cloud_volumes_ontap/networking_cloud_volumes_ontap.html on December 11, 2020.
Always check docs.netapp.com for the latest.

Table of Contents

Cloud Volumes ONTAP 1
Blogs 1

Cloud Volumes ONTAP

Blogs

NetApp Cloud Volumes ONTAP, A.K.A. ONTAP Cloud

Improved Performance Through Enhanced Networking Support

If you haven't heard the news, NetApp ONTAP Cloud has been rebranded as NetApp Cloud Volumes ONTAP. Along with the change comes exciting news of the release of support for ONTAP 9.4 – which in Amazon brings significant increases in front end performance. How cool is that, for once something really is both "New and Improved."

The most significant addition to the ONTAP 9.4 release for Cloud Volumes ONTAP is the inclusion of support for enhanced networking for the software defined storage system. As described by Amazon:

"Enhanced networking uses single root I/O virtualization (SR-IOV) to provide high-performance networking capabilities on supported instance types. SR-IOV is a method of device virtualization that provides higher I/O performance and lower CPU utilization when compared to traditional virtualized network interfaces. Enhanced networking provides higher bandwidth, higher packet per second (PPS) performance, and consistently lower inter-instance latencies. There is no additional charge for using enhanced networking."

Keep in mind that there are two types of enhanced networking interfaces supported by AWS and that NetApp supports at present the former (VF).

1. Intel 85299 Virtual Function (VF) interface which is supported within Xen hypervisor
2. Elastic network adapter supported from within KVM

The following graph calls out the increase in performance introduced by upgrading from ONTAP 9.3 to ONTAP 9.4 on an active-passive HA pair based on the c4.8xlarge instance type. Using the SLOB2 workload generator and Oracle 12.2c we performed a 75% read 25% update workload using an 8K database block size. While the read I/O rate increased 25% simply by upgrading an Cloud Volumes ONTAP cluster, the latency reduction is even more impressive.

[CV Networking support db file reads vs physical reads] |

CV_Networking_support_db_file_reads_vs_physical_reads.png

The upgrade path from ONTAP 9.3 to ONTAP 9.4 is a breeze with OnCommand Cloud Manager (OCCM). OCCM takes care of everything from upgrading the Cloud Volumes ONTAP nodes without downtime (assuming your environment is running in an HA configuration) to enabling sriov-net-support on the instances. Its as simple as clicking on the “New version available link” below and checking yes on a few “I have read and acknowledge the above.”

[CV Networking support cloud manager] | *CV_Networking_support_cloud_manager.png*

Not all instance types support enhanced networking, OCCM can help you there too by showing which instance types support this enchantment (those with the lightning bolt as shown below) as well as managing the process of instance type change in case you are interested in such. As the upgrade process, instance type change may be performed non-disruptively if running in an HA configuration.

[CV Networking support instance type] | *CV_Networking_support_instance_type.png*

About NetApp

NetApp is the data authority for hybrid cloud. We provide a full range of hybrid cloud data services that simplify management of data across cloud and on-premises environments to accelerate digital transformation. We empower global organizations to unleash the full potential of their data to expand customer touchpoints, foster greater innovation and optimize operations. For more information, visit: www.netapp.com #DataDriven

Copyright Information

Copyright © 2020 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.